Beyond the training set

Cross-domain AF detection with deep learning

Audited 12-lead ECG data, compact convolutional networks, calibrated inference and a Streamlit research dashboard.

97.07%
ACCURACY

88.32%
AF SENSITIVITY

99.65%
SPECIFICITY

0.9744
EXTERNAL AUROC

Georgia: 2,220 ECGs. Saved 95% bootstrap intervals: accuracy 96.31–97.75%; sensitivity 85.35–91.17%.

01 Background & objective

Atrial fibrillation (AF) is associated with stroke risk and may be unrecognized [1]. ECG models can encounter different patients, acquisition conditions and labels at a new site. We ask whether a model trained on two source domains retains discrimination and useful uncertainty on a third. The task is **AF versus explicit sinus rhythm (SR)**, not all-arrhythmia diagnosis.

02 Cohorts & data integrity

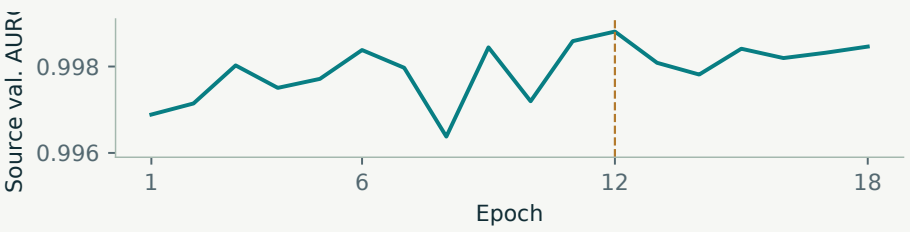
Domain	SR	AF	Role
PTB-XL	16,687	1,484	Source
Shaoxing	1,825	1,778	Source
Georgia	1,715	505	External

Of 24,027 label-eligible recordings, seven failed signal checks and 26 exact duplicates were removed, leaving **23,994 ECGs** [2,3]. Source partitions separate training (15,252), validation (2,173), temperature calibration (1,095), threshold calibration (1,078) and source testing (2,176).

03 Deep learning & evaluation protocol

Signals are ordered into 12 canonical leads, converted to mV and resampled to **100 Hz for 10 seconds**. Per-lead normalization uses training data only. Compact ResNet and InceptionTime-style CNNs are compared with empirical risk minimization, class-conditional feature alignment and CORAL [4,6]. Source validation selects ResNet with alignment weight 0.05. Three seeds (42, 202, 3407) form the exported ensemble.

12 × 1,000 input → 1D CNN → 128 features → AF logit



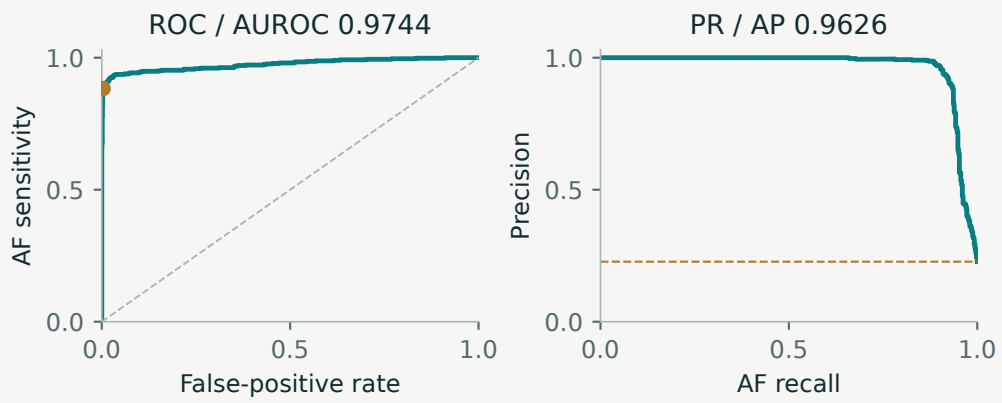
Training. AdamW, batch 256, cosine decay and early stopping; the selected seed-42 checkpoint is epoch 12.

04 Calibration & software implementation

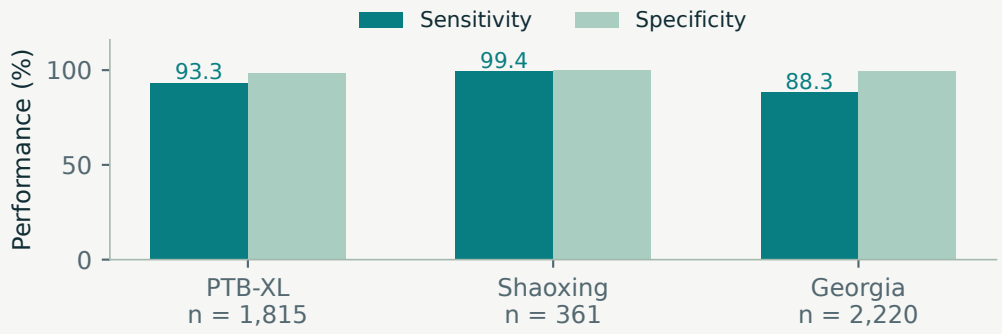
Source-only temperature scaling [5] precedes probability averaging. A frozen threshold of **0.6211** gives the AF/SR label; predictive entropy flags review. The source goal of 97% sensitivity at 95% specificity per domain was **not met** (minimum sensitivity: 95.31%).

The **Streamlit app** validates 100/500 Hz numerical ECG uploads and displays waveforms, predictions, model comparisons, calibration and robustness analyses. Thirteen software checks passed, including both exported ensembles and upload-to-result inference.

05 Results / Georgia external evaluation

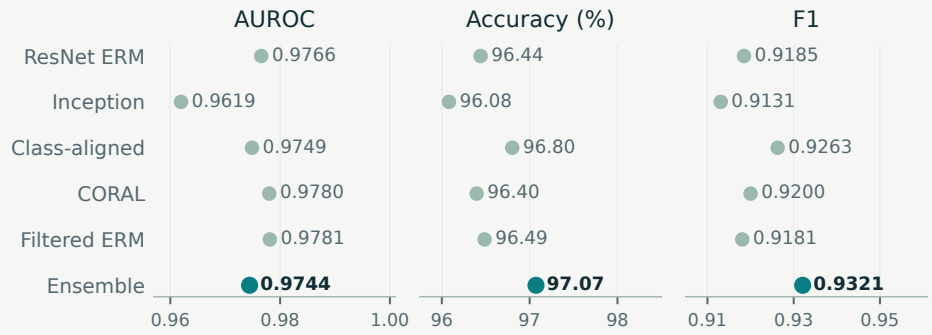


A / Discrimination. AUROC 0.9744; average precision 0.9626. The ROC marker is the frozen cutoff; the PR baseline is AF prevalence (22.75%).



B / Transfer. AF sensitivity falls from 93.29% (PTB-XL) and 99.44% (Shaoxing) to 88.32% (Georgia). Cohort differences prevent causal attribution.

06 Comparative results



C / Model comparison. Georgia results at each model's frozen source-selected cutoff. Axes are zoomed; values are labeled. Teal marks the selected ensemble.

07 Interpretation & limitations

Repeated Georgia evaluation is **exploratory** after prior-result inspection. Only one external rotation is complete; Shaoxing/Georgia IDs are patient proxies, and near duplicates may remain. Alignment does not dominate all metrics. Intervals condition on fitted models (200 cluster-bootstrap replicates). Independent cohorts and prospective workflow validation are required.

CONCLUSION

Strong external ranking and a working research application were achieved, but the AF sensitivity gap remains. The next step is independent validation, not a claim of clinical readiness.

SELECTED REFERENCES / sources and method foundations

[1] CDC. About Atrial Fibrillation (2024).
[2] Wagner et al. PTB-XL, v1.0.3. PhysioNet (2022).
[3] PhysioNet. Challenge 2021, v1.0.3; cohort provenance.
[4] Sun & Saenko. Deep CORAL (2016).
[5] Guo et al. On Calibration of Modern Neural Networks. ICML (2017).
[6] Fawaz et al. InceptionTime. DMKD (2020).

Latest run: 20261027_205834. Retained-cohort counts and plots use saved project artifacts. Research demonstration; no clinical utility claim.